

Program: Diploma in Automation and Robotics	
Course Code: 4339	Course Title: Industry 4.0
Semester: 4	Credits: 0
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To learn and understand basics of Industry 4.0
- To develop knowledge on the Smart Manufacturing Solutions
- To identify new outlook of opportunities of smart industry
- Acquire skill to face the future challenges and opportunities smart Industry

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic principles and theorms of Engineering Physics		Engineering Physics	1
Basic Knowledge of Sensors and actuators		Foundations of Robotics	2
Computer Skills		Python Programming	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basic concepts of Industry 4.0 s	11	Understanding
CO2	Understand Industry4.0 and different technologies and learn PLC programming	11	Applying
CO3	Understandbuilding blocks of industry 4.0	11	Understanding
CO4	Explore and Understand advances in robotics in the era of industry 4.0.	10	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the basic concept of Industry 4.0		
M1.01	Understand the origin and concept of industry 4.0	1	Understanding
M1.02	Understand basic components of industry 4.0	2	Understanding
M1.03	Understand the supportive technologies	4	Understanding
M1.04	Understand proposed framework of industry 4.0	2	Understanding
M1.05	Prepare a model sketch of a smart industry	2	Applying
Contents: Introduction to Industry 4.0: Introduction, Core idea of Industry 4.0, Origin concept of industry 4.0 - A Conceptual Framework for Industry 4.0: Introduction, Main Concepts and Components of Industry 4.0, State of Art, Supportive Technologies, Proposed Framework for Industry 4.0.			
CO2	Understand Industry 4.0 and different Technologies and PLC programming		
M2.01	Understand the human machine interactions virtual reality and augmented reality systems	2	Understanding.
M2.02	Understand Advanced robotics and 3-D printing	2	Understanding.
M2.03	Understand Cloud Computing and Concept of “Equipment-As-a-Service,” EAAS - Connectivity Solutions	2	Understanding.
M2.04	Study the basics and applications of Bluetooth, BLE, Bluetooth 5.0, ZigBee, ZigBee 3.0, Z-Wave, 6LowPAN, RFID, WiFi,	2	Understanding.

	Mobile/Cellular, SATCOM		
M2.05	Learn PLC, PLC Programming (Ladder Logic Programming)	3	Applying
	Series Test – I	1	
Contents: Industry 4.0 and Technologies- Human-machine interaction - Touch interfaces -Virtual reality and Augmented-reality systems - Transferring digital instructions to the physical world - Advanced robotics and 3-D printing - Autonomous activities - Lean Manufacturing -,Cloud Computing and Concept of “Equipment-As-a-Service,” EAAS - Connectivity Solutions: Bluetooth, BLE, Bluetooth 5.0, ZigBee, ZigBee 3.0, Z-Wave, 6LowPAN, RFID, WiFi, Mobile/Cellular, SATCOM, PLC, PLC Programming (Ladder Logic Programming) - Proactive/Predictive Maintenance and Continuous Monitoring			
CO3	Understand Building Blocks of Industry 4.0		
M3.01	Understand Building blocks of industry 4.0, Cyber physical systems	2	Understanding
M3.02	Key characteristics of cyber physical systems	1	Understanding
M3.03	Familiarizethe concept of IIOT	2	Applying
M3.04	Understand the horizontal and vertical integration and RAMI 4.0	2	Understanding
M3.05	Application of Internet of things in smart applications	2	Applying
M3.06	Familiarize IoT and Industry 4.0 Connectivity	2	Understanding
Contents: Building Blocks of Industry 4.0: Cyber Physical Systems - Cyber-physical systems (CPS) in the Industry 4.0 vision - Cyber-physical systems key characteristics - Industry 4.0 building blocks - The (Industrial) Internet of Things - Industry 4.0 principles: horizontal and vertical integration - The Reference Architectural Model Industry 4.0 (RAMI 4.0) - Internet of Things use cases - Smart factories, smart plants and smart applications - Internet of Things and Industry 4.0 connectivity - CPS-enabled capabilities			
CO4	Explore and Understand advances in robotics in the era of industry 4.0		
M4.01	Explore the advances in robotics in the era of industry 4.0 introduction, recent components of Robots	2	Understanding
M4.02	Understand advanced sensor technologies	2	Understanding
M4.03	Familiarize with Internet of Robotic Things, Cloud Robotics	2	Understanding
M4.04	Understand the Cognitive Architecture for Cyber-Physical Robotics	2	Applying

M4.05	Industrial Robotic Applications	2	Understanding
	Series Test – II	1	
Contents: Advances in Robotics in the Era of Industry 4.0: Introduction, Recent Technological Components of Robots- Advanced Sensor Technologies, Internet of Robotic Things, Cloud Robotics, and Cognitive Architecture for Cyber-Physical Robotics, Industrial Robotic Applications- Manufacturing, Maintenance and Assembly.			

Text / Reference

T/R	Book Title/Author
T1	Handbook of Industry 4.0 and SMART Systems, by Diego Galar Pascual (Author), Pasquale Daponte (Author), Uday Kumar (Author)
R1	Alp Ustundag and Emre Cevikcan, "Industry 4.0: Managing the Digital Transformation".
R2	Bartodziej, Christoph Jan, "The Concept Industry 4.0".
R3	Klaus Schwab, "The Fourth Industrial Revolution"
R4	Randy Frank, "Understanding Smart Sensors", Artech House, Second Edition, 2011 Boston,
R5	Industry 4.0: The Industrial Internet of Things", by Alasdair Gilchrist (Apress), 2017

Online Resources

Sl No	Website Link
1	https://nptel.ac.in/courses/106/105/106105195/
2	https://www.udemy.com/course/intro-to-industry-4/